

Program 10

Demonstrate minikube dashboard using CLI and Kubernetes Cluster

Program Structure:

```
program10/  
----->deployment.yaml  
----->services.yaml
```

Procedure:

step 1: start minikube
minikube start
check the status of minikube
minikube status

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~$ mkdir prg10  
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~$ cd prg10  
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/prg10$ minikube start --cpus=2 --memory=4096  
🐳 minikube v1.37.0 on Ubuntu 22.04  
🔧 Using the docker driver based on existing profile  
⚠️ You cannot change the memory size for an existing minikube cluster. Please first delete the cluster.  
👉 Starting "minikube" primary control-plane node in "minikube" cluster  
📦 Pulling base image v0.0.48 ...  
🔄 Restarting existing docker container for "minikube" ...  
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...  
🔍 Verifying Kubernetes components...  
■ Using image gcr.io/k8s-minikube/storage-provisioner:v5  
🌟 Enabled addons: storage-provisioner, default-storageclass  
  
❗ /usr/bin/kubectl is version 1.30.14, which may have incompatibilities with Kubernetes 1.34.0.  
■ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'  
🏁 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default  
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/prg10$ minikube status  
minikube  
type: Control Plane  
host: Running  
kubelet: Running  
apiserver: Running  
kubeconfig: Configured  
  
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/prg10$ minikube dashboard  
🔧 Enabling dashboard ...  
■ Using image docker.io/kubernetesui/dashboard:v2.7.0  
■ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8  
💡 Some dashboard features require the metrics-server addon. To enable all features please run:  
  
minikube addons enable metrics-server  
  
🐼 Verifying dashboard health ...  
🚀 Launching proxy ...  
🐼 Verifying proxy health ...  
🌐 Opening http://127.0.0.1:37061/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser  
Opening in existing browser session.  
^C
```

Step 2: Create deployment.yaml file with the following code
nano deployment.yaml

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx: ~/prg10
GNU nano 6.2 deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

step 3: Create service.yaml file with the following code
nano service.yaml

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx: ~/prg10
GNU nano 6.2 service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

step 4 : Apply the file

- kubectl apply -f deployment.yaml
- kubectl apply -f service.yaml
- Get pods,deployment,services
- kubectl get pods
- kubectl get deployment
- kubectl get service

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/prg10$ kubectl apply -f deployment.yaml
kubectl apply -f service.yaml
deployment.apps/nginx-deployment unchanged
service/nginx-service unchanged
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/prg10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-6k6vn   1/1     Running   1 (2m18s ago)    19m
nginx-deployment-6f9664446b-dqwfc   1/1     Running   1 (2m18s ago)    19m
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/prg10$ kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    2/2     2             2            19m
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/prg10$ kubectl get services
NAME                TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1        <none>         443/TCP           13d
nginx-service        NodePort    10.103.181.149   <none>         80:30249/TCP      19m
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/prg10$ kubectl describe nginx-deployment-6f9664446b-6k6vn
error: the server doesn't have a resource type "nginx-deployment-6f9664446b-6k6vn"
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/prg10$ kubectl describe <nginx-deployment-6f9664446b-6k6vn>
bash: syntax error near unexpected token `newline'
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/prg10$ kubectl describe pod nginx-deployment-6f9664446b-6k6vn
Name:                nginx-deployment-6f9664446b-6k6vn
Namespace:           default
Priority:             0
Service Account:     default
Node:                minikube/192.168.49.2
Start Time:          Wed, 31 Dec 2025 10:28:45 +0530
Labels:              app=nginx
```

step 5: Check the minikube service

minikube service nginx-service

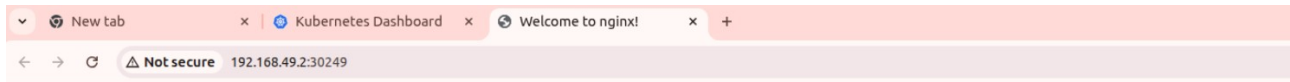
```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/prg10$ minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:30249

🌐 Opening service default/nginx-service in default browser...

```
1RV24MC073_pavithra@pavithra-HP-Laptop-15-db1xxx:~/prg10$ minikube service nginx-service
```

Out put



Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.